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Deciphering the Roles of MA-Based Volatile Additives for α-$FAPbI_3$ to Enable Efficient Inverted Perovskite Solar Cells

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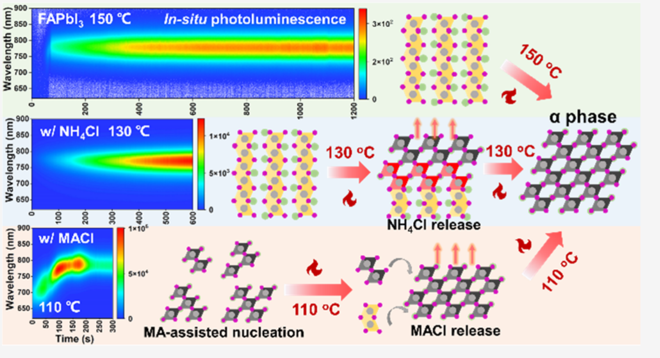
ABSTRACT: Functional additives that can interact with the perovskite precursors to form the intermediate phase have been proven essential in obtaining uniform and stable α-$FAPbI_3$ films. Among them, Cl-based volatile additives are the most prevalent in the literature. However, their exact role is still unclear, especially in inverted perovskite solar cells (PSCs). In this work, we have systematically studied the functions of Cl-based volatile additives and MA-based additives in formamidinium lead iodide ($FAPbI_3$)-based inverted PSCs. Using in situ photoluminescence, we provide clear evidence to unravel the different roles of volatile additives (NH_4Cl, FACl, and MACl) and MA-based additives (MACl, MABr, and MAI) in the nucleation, crystallization, and phase transition of $FAPbI_3$. Three different kinds of crystallization routes are proposed based on the above additives. The non-MA volatile additives (NH_4Cl and FACl) were found to promote crystallization and lower the phase-transition temperatures. The MA-based additives could quickly induce MA-rich nuclei to form pure α-phase $FAPbI_3$ and dramatically reduce phase-transition temperatures. Furthermore, volatile MACl provides a unique effect on promoting the growth of secondary crystallization during annealing. The optimized solar cells with MACl can achieve an efficiency of 23.1%, which is the highest in inverted $FAPbI_3$-based PSCs.



■ INTRODUCTION

Perovskite solar cells (PSCs) have emerged as a promising technology for the photovoltaic industry owing to their overwhelmingly improved performance and the cost-effective solution processing process.[1−9] Although inverted (*p*−*i*−*n* structure) PSCs are attractive for future commercialization due to their low-temperature processability and better compatibility with flexible substrates and perovskite-based tandem device fabrication,[10−15] they typically exhibit lower power conversion efficiency (PCE) than their regular (*n*−*i*−*p*) counterparts.[16−18] Most certified record PCEs are based on single-junction *n*−*i*−*p* PSCs using an optimized formamidinium lead triiodide ($FAPbI_3$) perovskite absorber layer.[19−21] However, the excellent performance of $FAPbI_3$ in *n*−*i*−*p* PSCs could not be directly translated into efficient inverted PSCs. So far, only a few studies have reported using $FAPbI_3$ in inverted PSCs, and the PCEs are usually below 22%.[22,23] The main bottleneck limiting the PCE of $FAPbI_3$-based inverted PSCs is the needed high-temperature phase transition ($\geq$150 °C) for forming α-phase $FAPbI_3$, which is detrimental to inverted PSCs.[23]

To overcome this problem, a widely used strategy in inverted PSCs is compositional engineering by alloying $FAPbI_3$ with a small fraction of non-native cations, such as Cs^+, MA^+, and Rb^+ cations, and a Br^- anion, as used in most state-of-the-art PSCs.[10−12,24,25] However, tuning of compositions will inevitably increase the bandgap and suffer from possible local phase segregation of non-native cations and halides.[26] Therefore, instead of using a substitution strategy, volatile additives have been incorporated in the perovskite precursor solution to form a pure black-phase α-$FAPbI_3$ perovskite film and increase crystallinity.[21,27,28]

To facilitate the crystallization and phase transition at relatively low temperatures, the additives are expected to coordinate with the Pb atom in the precursor and be easily released during the film annealing process.[26] In this regard, small-volume cations with amino groups are good candidates for volatile additives. Besides, chloride additives have been proven to effectively promote perovskite crystallization, which is conducive to perovskite morphology and film quality.[29,30] Therefore, several volatile additives such as ammonium

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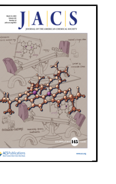

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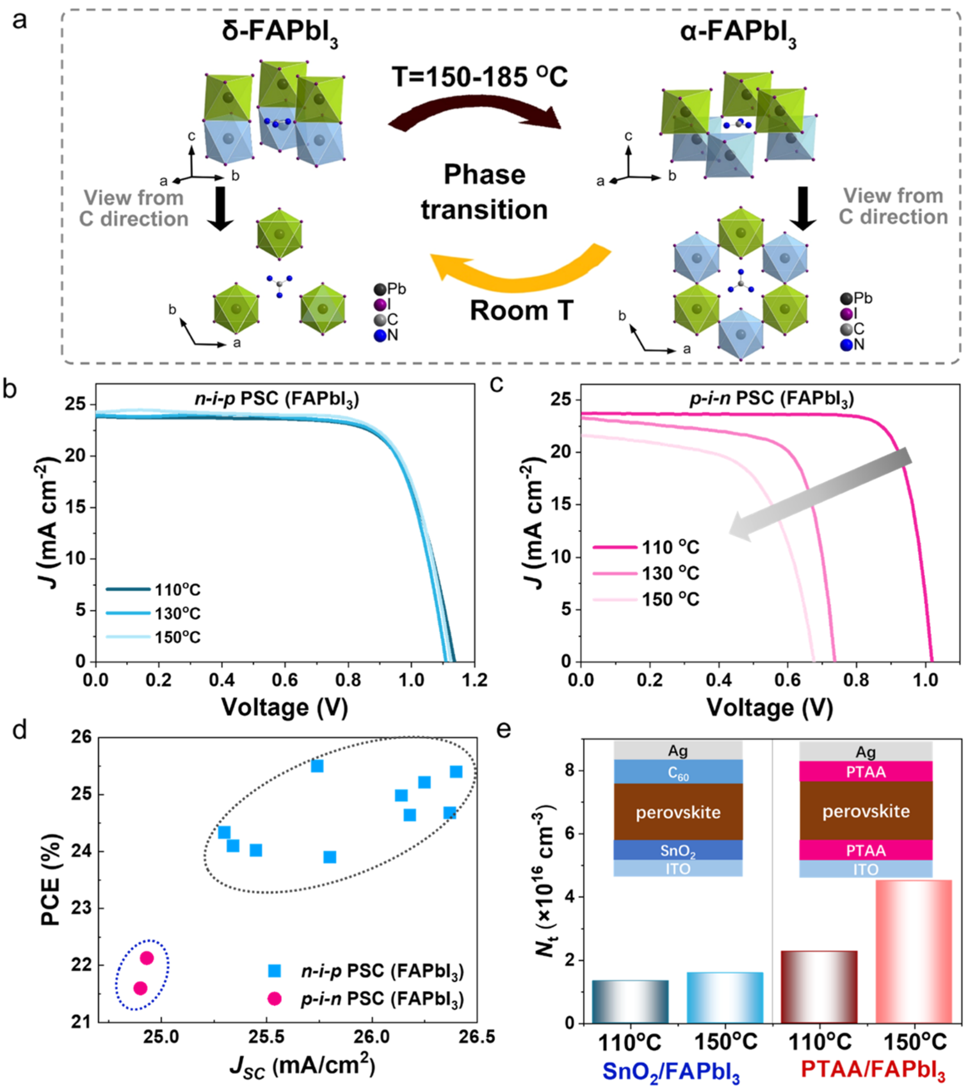

Figure 1. (a) Crystalline structures viewed from different directions showing δ-$FAPbI_3$ and α-$FAPbI_3$ phases and the phase transition between them. Three $[PbI_6]$ octahedra in the upper or lower layer are labeled green or blue, respectively. The view of δ-$FAPbI_3$ from the C direction shows two overlapped layers, while in the case of α-$FAPbI_3$, it shows crossed structure between two layers. Pb, I, C, and N atoms are labeled deep gray, purple, gray, and blue, respectively. (b, c) J−V curve evolution of (b) n−i−p and (c) p−i−n structured PSCs with the different annealing temperatures. (d) Recent advances in efficient $FAPbI_3$-based PSCs in n−i−p or p−i−n structures. (e) Defect density from space charge-limited current (SCLC) measurement of $FAPbI_3$ based on SnO_2 or PTAA substrates annealed at different temperatures.

chloride (NH_4Cl),[31,32] methylamine chloride (MACl),[21,33] and formamidine chloride (FACl)[28] have been applied to facilitate the crystallization and phase transition of α-$FAPbI_3$. However, the guideline for choosing the most suitable additive and understanding how additives affect the nucleation and crystallization of α-$FAPbI_3$ is still unclear.[34]

Therefore, we have systematically studied the roles of volatile additives (NH_4Cl, FACl, and MACl) and MA-based additives (MACl, MABr, and MAI) in the nucleation, crystallization, and phase transition of $FAPbI_3$ using in situ photoluminescence (PL). This work revealed three different kinds of $FAPbI_3$ crystallization routes, which could provide helpful guidance for choosing appropriate additives to achieve high-performance inverted PSCs. We found that non-MA volatile additives (NH_4Cl and FACl) could promote crystallization and lower the phase-transition temperatures. MA-based additives could quickly induce MA-rich nuclei to facilitate the formation of α-phase $FAPbI_3$ and dramatically reduce phase-transition temperatures. Furthermore, MACl could combine benefits from volatile and MA-containing additives to promote secondary crystallization during annealing, as revealed in the in situ PL. With the above understanding, the optimized solar cells with MACl could be fabricated to achieve 23.1% efficiency, which is one of the best in $FAPbI_3$-based inverted PSCs.

■ RESULTS AND DISCUSSION

As shown in Figure 1a, the photoactive black α-phase $FAPbI_3$ perovskite is thermodynamically unstable, which will spontaneously transform into a photo-inactive δ-phase at room temperature, especially in humid ambient air.[35] During the crystallization of $FAPbI_3$, α- and δ-phase $FAPbI_3$ are formed simultaneously, and δ-$FAPbI_3$ will undergo a reversible phase transition to α-$FAPbI_3$ when the temperature rises above 150 °C.[35] To gain a thorough understanding of the lattice and ionic changes between δ-$FAPbI_3$ and α-$FAPbI_3$ during the reversible phase transition, the single-crystal structure data of δ-$FAPbI_3$ and α-$FAPbI_3$ were further analyzed. Figure 1a shows different

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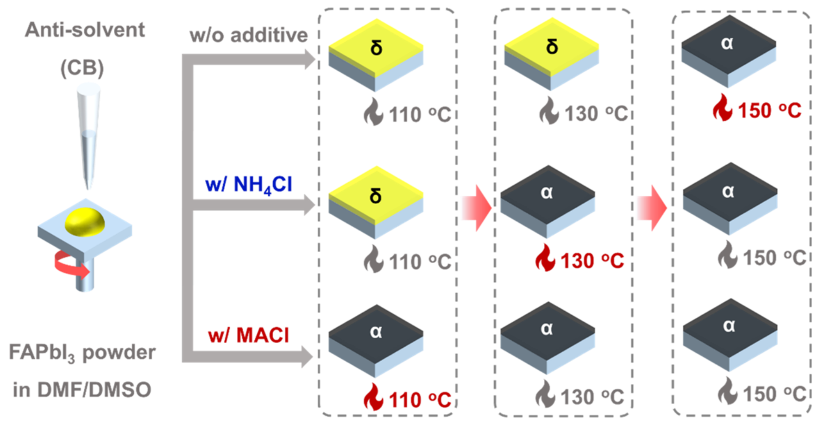
Scheme 1. Schematic Diagram of $FAPbI_3$ Films for Control and with NH_4Cl or MACl under Different Annealing Temperatures



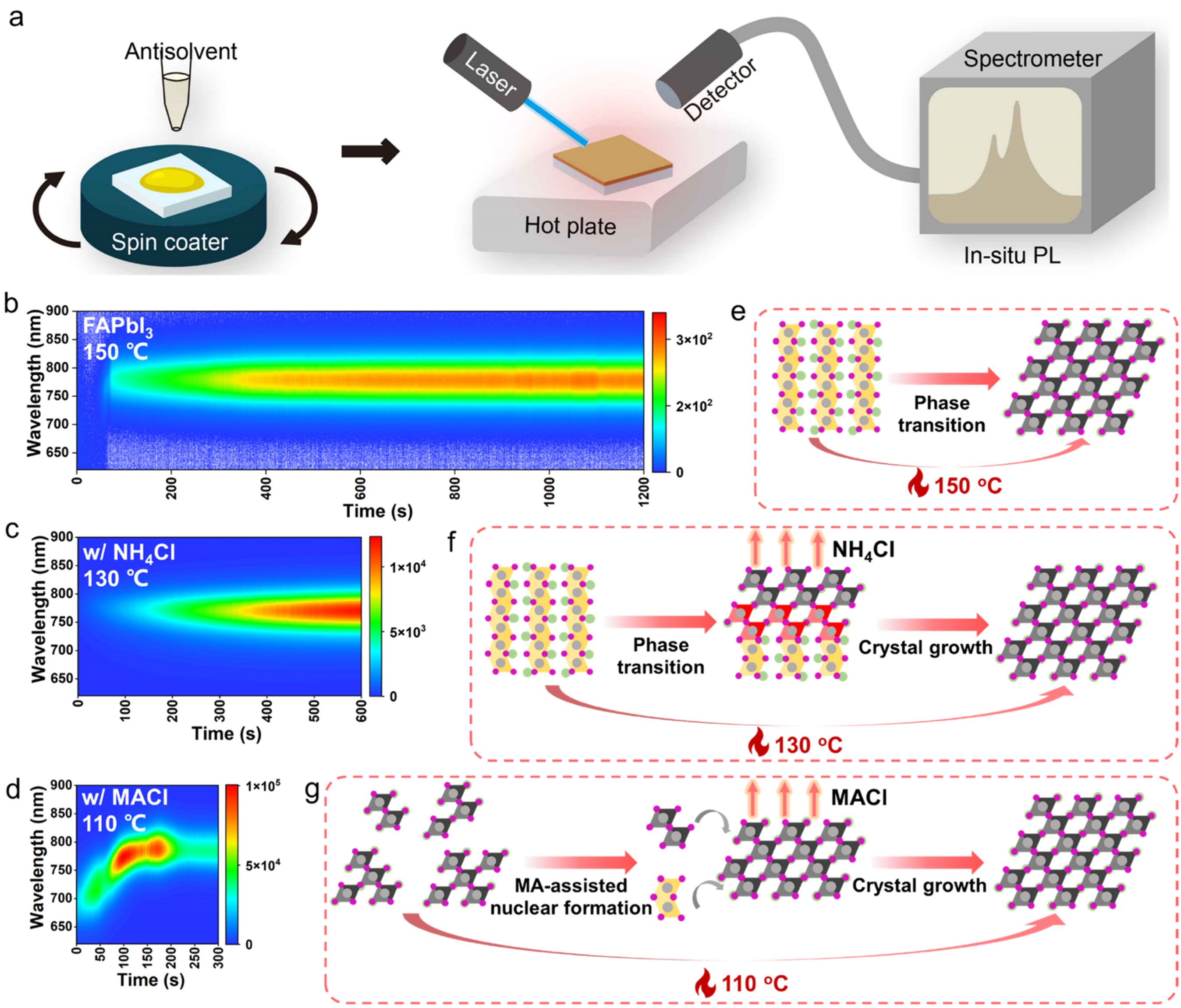


Figure 2. (a) Schematic illustration of in situ photoluminescence (PL) monitoring during the $FAPbI_3$ perovskite crystallization. (b−d) Heat maps of in situ PL of the $FAPbI_3$ film under the different annealing temperatures for (b) control, (c) with NH_4Cl, and (d) with MACl. (e−g) Schematic illustration for crystallization kinetics of different phase transition and crystallization routes to the α-$FAPbI_3$ film through thermal annealing or volatile additive-assisted crystallization under the various annealing temperatures for (e) control, (f) with NH_4Cl, and (g) with MACl.

views of the lattice structure in δ-$FAPbI_3$ and α-$FAPbI_3$, suggesting that the phase transition between δ-$FAPbI_3$ and α-$FAPbI_3$ may arise from the movement of ions between adjacent layers.

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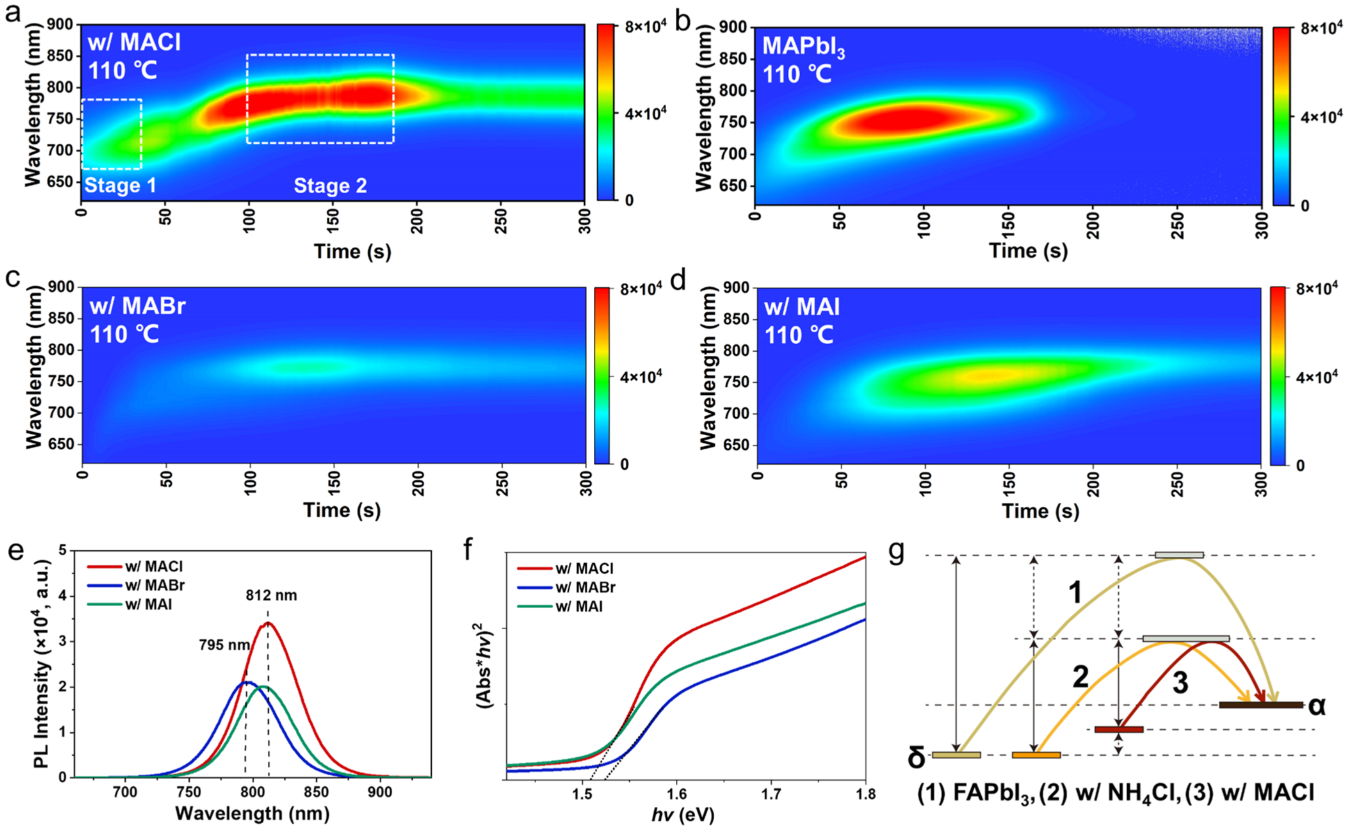

Figure 3. (a–d) Heat maps of in situ PL for (a) $FAPbI_3$ with MACl, (b) $MAPbI_3$, (c) $FAPbI_3$ with MABr, and (d) $FAPbI_3$ with MAI under 110 °C annealing temperature. (e) PL spectra of $FAPbI_3$ films with the MACl, MABr, or MAI additive. (f) Tauc plot of $FAPbI_3$ films with the MACl, MABr, or MAI additive. (g) Schematic free energy for forming $FAPbI_3$ perovskites for control, with the NH_4Cl or MACl additive.

Detailed analysis and discussion are shown in Figures S1–S3. Encouragingly, significant progress has been made toward stabilizing α-$FAPbI_3$ without changing the bandgap, which significantly led to the recent ascending of PCE for $FAPbI_3$-based regular PSCs over 25%[16,17,19−21,27,36−40] (Figure 1d). However, the PCE of $FAPbI_3$-based inverted PSCs was still less than 22%[22,23,41] (Figure 1d), far behind that of the regular PSCs. The regular and inverted-structured PSCs adopt different substrate materials for perovskite deposition, which might be a critical reason for their remarkably different photovoltaic performances. Therefore, we compared regular and inverted-structured PSCs based on the one-step deposition of $FAPbI_3$ films to study their potential variations. The details on device fabrication can be seen in the Supporting Information. We found that the annealing process of $FAPbI_3$ perovskite films on HTL substrates significantly differed from that of regular PSCs. According to the literature, the optimal annealing condition of the $FAPbI_3$ perovskite on inorganic electron-transporting layers like TiO_2 or SnO_2 is heating at 150 °C for 10–20 min. However, the same fabrication condition resulted in deficient photovoltaic performance for inverted-structured PSCs (Figure 1b,c).

Therefore, we have systematically optimized the annealing conditions by decreasing the annealing temperature from 150 to 130 to 110 °C. The detailed photovoltaic parameters are compared in Table S2. Intriguingly, we found that the photovoltaic performance of $FAPbI_3$-based inverted PSCs could be improved upon decreasing the annealing temperature of perovskite films, and the best results were obtained at 110 °C. However, the performance of the regular PSC using different annealing temperatures remained unchanged. To clarify the considerable difference, we studied the morphology and crystallinity of $FAPbI_3$ films on PTAA and SnO_2 annealed at 110 and 150 °C, respectively. From the X-ray diffraction (XRD) and scanning electron microscopy (SEM) measurements (Figures S5 and S6), the PTAA substrates seemed not to change the properties of $FAPbI_3$ perovskite films at high annealing temperatures compared with SnO_2.

The trap density of $FAPbI_3$ films based on PTAA or SnO_2 was further evaluated by the space charge-limited current (SCLC) measurements based on hole-only devices with the structure of ITO/PTAA/$FAPbI_3$/PTAA/Ag or electron-only devices with the structure of ITO/ SnO_2/$FAPbI_3$/C60/Ag.[42,43] As shown in Figures 1e and S7, the trap density of $FAPbI_3$ coated on SnO_2 was 1.34×10^{16} and 1.56×10^{16} cm^{-3} for samples after being annealed at 110 and 150 °C, respectively, which only shows a slightly increased trap density. However, the $FAPbI_3$ coated on PTAA annealed at 150 and 110 °C showed higher trap densities of 2.27×10^{16} and 4.51×10^{16} cm^{-3}. The gap between them is much larger than that for the SnO_2 substrate, indicating that the annealing process at 150 °C might promote more interface defects in inverted devices. NiO_x and SAM-based substrates also show a similar increased defect density at 150 °C as shown in Figure S7. Therefore, effectively reducing the temperature of phase transition and thermal annealing to achieve α-$FAPbI_3$ may be the key to improving the PCE of $FAPbI_3$-based inverted PSCs.

Additive engineering is a common strategy for controlling the crystal phase and surface morphology of perovskite

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films.[21,26] To study the underlying mechanism of how additives affect the crystallization of α-$FAPbI_3$ and guide for choosing appropriate additives, NH_4Cl and MACl were selected as the main volatile additives to study their effects on the $FAPbI_3$ crystallization and phase transition. The precursor solution was prepared by dissolving selected additives and $FAPbI_3$ powder into a DMF/DMSO mixed solvent. As shown in Scheme 1, Figures S8, and S9, photos of perovskite films were taken after annealing at different temperatures, where the phase transition was further confirmed by XRD measurements. We found that the film with NH_4Cl had a lower phase-transition temperature (130 °C) than that without additives (150 °C). Moreover, the film with MACl could achieve the phase transition below 110 °C (even at 60 °C), which indicated that MACl could more effectively promote the formation of α-$FAPbI_3$.

To further investigate the roles of additives in $FAPbI_3$ formation, in situ photoluminescence (PL) measurements were conducted to monitor the $FAPbI_3$ perovskite crystallization process as shown in Figure 2a. After the precursor solution was spin-coated onto a glass substrate, the sample was immediately transferred to a hot plate and monitored by in situ PL. The annealing process was performed under different temperatures according to the phase-transition temperature described in Scheme 1. Figures 2b−d, S11, S12, and S14 show the heat maps of the PL spectra for $FAPbI_3$, w/NH_4Cl, and w/MACl taken during the annealing process.

The shifts of the PL peaks in Figures 2b and S11 showed that the $FAPbI_3$ sample reached its maximum PL intensity of 3×10^2 when heated at 150 °C for 150 s. The introduction of NH_4Cl reduced the phase-transition temperature to below 130 °C and enhanced the maximum PL intensity by 2 orders of magnitude to 1.3×10^4, indicating the improved crystallinity and film quality of $FAPbI_3$. Note that volatile FACl also had a similar effect to NH_4Cl, as shown in Figure S10. Unlike the first two crystallization routes, the sample with an MA-based volatile additive MACl had a faster crystallization rate. It reached the maximum PL intensity (1×10^5), which was one more order of magnitude higher than that for the NH_4Cl sample, showing a further increased crystallization rate and film crystallinity. Moreover, different from NH_4Cl and FACl, the maximum PL peak position of the MACl-based sample gradually shifted from 700 to 780 nm during the first 100 s of annealing. This significant peak shift may be attributed to MACl-assisted nucleation, which dramatically reduced the phase-transition temperature and simultaneously increased the speed of forming α-phase $FAPbI_3$. Based on our observations, we show a schematic diagram of possible crystallization kinetics, as shown in Figure 2e−g.

To investigate the unique PL maximum peak shift caused by the addition of MACl during heating, the $MAPbI_3$ sample was tested during annealing by in situ PL (Figures 3b and S15). We found that the $MAPbI_3$ sample also showed a similar red shift of the maximum PL peak in the first 100 s of annealing, gradually moving from 700 to 750 nm. Furthermore, the maximum PL peak intensity also reached 1×10^5. This phenomenon shows that MA should be the dominating component in the $FAPbI_3$ sample with the MACl additive in the first phase transition and crystallization stage to form an MA-based intermediate nucleus to induce faster crystallization.

The perovskite precursor solution was characterized by dynamic light scattering (DLS) to verify the MA-assisted nucleation process (Figure S17). We found that both peaks with particle sizes between 1−10 and 1−10 μm in the precursor solution shifted to larger sizes as the concentration of MACl increased from 0 to 40 mol %, indicating that MACl promotes the formation and growth of crystal nuclei in the perovskite precursor solution. Furthermore, when the annealing time ranges from 30 to 300 s, the characteristic peak of the α phase in the XRD pattern (Figure S18) moves from 14.08 to 13.97°, indicating that chloride and MA were substituted by iodide and FA during the annealing process.[21]

To further verify our speculation, the effect of MA-based nonvolatile additives (MABr and MAI) was investigated on the phase transition of $FAPbI_3$ (Figures 3c,d, S19, and S20). The PL results showed that MABr and MAI as additives also caused a red shift of the PL maximum peak like that of $MAPbI_3$ and MACl samples, which further proved that MA-based additives could play a role in assisting nucleation in the initial stage of $FAPbI_3$ film formation and phase transition. All four samples in Figure 3a−d underwent an initially increased then decreased PL intensity. The changes in PL intensity result from the balance between radiative and nonradiative recombination. In the first stage of annealing, the volatilization of the solution in the precursor film leads to supersaturation of the solute. Then, many luminating nanocrystals are formed at this stage, resulting in a gradually increased PL intensity. In the second annealing stage, there are three possible reasons for the decreased PL intensity: (1) during the growth and fusion of tiny crystals, defects are formed in grains and grain boundaries to cause significant nonradiative recombination.[44] (2) Continuous heating will form thermally activated carrier traps within the perovskite to increase nonradiative recombination.[45,46] (3) Continuous heating will also cause lattice vibration to increase the loss of excitation energy of the luminescence center.[47] As a result, the PL intensity started to decrease after reaching the peak. Further comparing the difference between volatile MACl and nonvolatile MABr and MAI, we found that, when the MACl sample film was heated to 100−170 s, the maximum PL peak signal height first decreased and then increased, while this phenomenon did not exist in $FAPbI_3$ samples added with MAI or MABr and $MAPbI_3$ samples. We marked the dotted square in Figure 3a as stage 2. At this stage, the PL peak intensity decreased slightly after reaching the maximum peak intensity like other samples. Subsequently, the maximum PL peak intensity in the MACl sample was enhanced again, indicating that it can promote the secondary crystal growth of the perovskite film due to the rapid release of volatile MACl. To verify the effect of secondary crystal growth, we have monitored the changes of the w/MACl precursor film under different annealing times by SEM. The SEM results are shown in Figure S22a−f. We found that the w/MACl precursor film formed grains of 0.3−1 μm after being annealed for 30 s. With the prolonged annealing time from 30 to 180 s, the tiny grains gradually merged and the grain sizes gradually increased. During annealing for 90−180 s, an obvious growth of the crystals was observed, which corresponds to stage 2 in the in situ PL in Figure 3a. However, when heated to 300 s, excess lead iodide appeared on the perovskite surface, which may be beneficial to passivate surface defects. This phenomenon did not appear in the nonvolatile MABr and MAI samples, which once again proved that volatile MACl had a unique auxiliary nucleation and could promote the secondary crystal growth of perovskite films to obtain better $FAPbI_3$ films.

As shown in Figure 3e, the PL maximum peak of the obtained MACl sample was 3 nm red-shifted than that of the MAI sample and 17 nm red-shifted than that of the MABr

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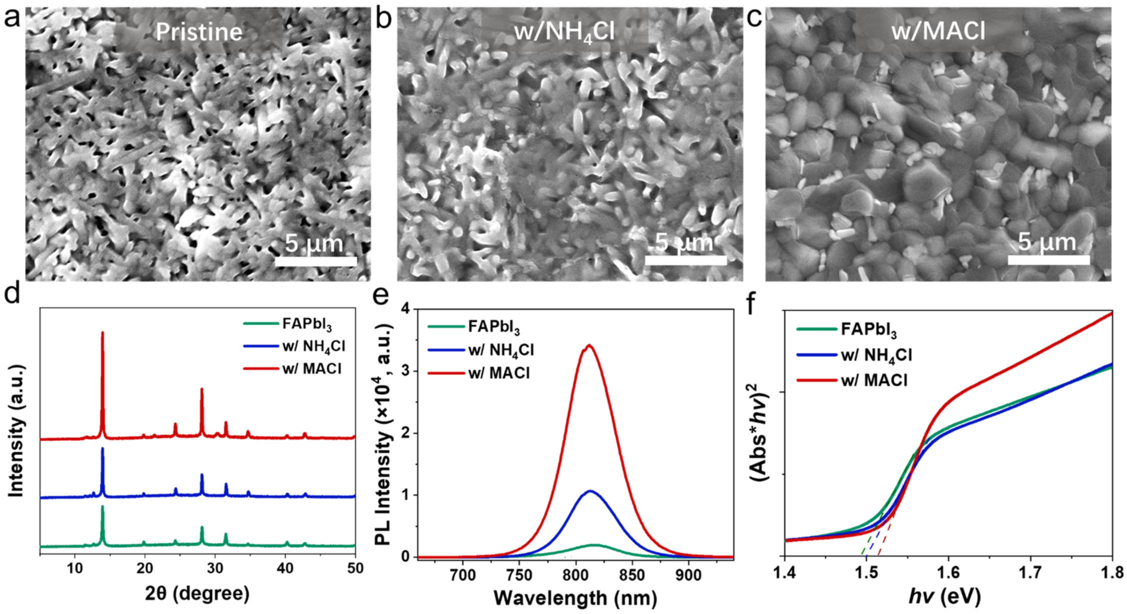

Figure 4. (a−c) Top-view SEM images of the $FAPbI_3$ films for (a) control, (b) with NH_4Cl, and (c) with MACl. (d) The XRD patterns of $FAPbI_3$ films for control, with NH_4Cl, and with MACl. (e) PL spectra of $FAPbI_3$ films for control, with NH_4Cl, and with MACl. (f) Tauc plot of $FAPbI_3$ films for control, with NH_4Cl, and with MACl.

sample. These results suggested that the volatility of MACl could maintain the narrowest bandgap while lowering the $FAPbI_3$ phase-transition temperature and promoting crystallization. The narrowest bandgap and higher absorption coefficient from the Tauc plot corresponding to the ultraviolet−visible (UV−Vis) absorption spectrum in Figure 3f were also consistent with the PL results.

In summary, the MA-based additives formed a mesophase-like $MAPbI_3$ perovskite during the crystallization process, which on the one hand increased the nucleation number; on the other hand, it also decreased the phase-transition temperature to below 110 °C. In addition, the volatile MA-based additive (MACl) could also promote the secondary crystal growth of perovskite films, which was very significant for obtaining high-performance $FAPbI_3$-based inverted devices. As shown in Figure 3g, the results discussed above could be attributed to three different crystallization pathways for α-$FAPbI_3$. The first route is that the additive-free pristine $FAPbI_3$ achieves phase transition by pure internal energy, so 150 °C is required to increase the internal energy to activate the Pb and I atoms through the energy barrier. Route 2 shows a lower energy barrier due to volatile additives, taking NH_4Cl as an example. The incorporation of NH_4Cl in the precursor can decrease the δ to α phase-transition temperature to below 130 °C. This phenomenon may originate from the principle like the liquid−solid interface[48−50] and template growth.[51] In other words, NH_4Cl may coordinate with Pb and be released to activate atomic movement, thereby reducing the internal energy required to achieve phase transition at 130 °C. Route 3 represents the use of MA-based volatile additives, exemplified by MACl. The introduction of MACl can form an MA-dominated intermediate core. Compared with route 2, the initial energy level of route 3 is higher, so lower internal energy (<110 °C) can help to cross the energy barrier to promote phase transition.

Scanning electron microscopy (SEM) images were obtained to study the morphology of $FAPbI_3$ films of three different crystallization routes. As shown in Figure 4a, the pristine $FAPbI_3$ film showed an average domain size of about 100 nm and some black spots, as further verified by AFM in Figure S23. However, the perovskite film had fewer pinholes with the incorporation of NH_4Cl (Figure 4b). The film became more uniform with increased grain sizes (200−400 nm), probably due to the $FAPbI_3$ growth promoted by the volatile additive. The MACl additive could also significantly increase the grain sizes to be larger than 1 μm (Figure 4c). The AFM measurements also verified that adding NH_4Cl and MACl additives significantly increased the grain sizes. Note that a similar phenomenon was observed in the case of $FAPbI_3$ with MABr or MAI. As shown in Figure S18, the $FAPbI_3$ with MABr or MAI presented a smoother surface and larger grain sizes in SEM images. The improved morphology of the perovskite with MA-based additives probably originated from the MA-rich nuclei that could regulate the crystal and growth of $FAPbI_3$ thin films.[21,26]

X-ray diffraction (XRD) measurements were performed to study the effect of additives on the crystallinity of $FAPbI_3$ perovskite films. As illustrated in Figure 4d, the control film without additives showed relatively weak diffraction peaks, suggesting its poor crystallinity. Interestingly, the diffraction peak intensity was enhanced for $FAPbI_3$ films with MACl additives. The peak intensity of the (001) crystallographic plane at 14.0° in the target film with MACl was approximately 2 times higher than that of the control $FAPbI_3$ film, indicating its increased crystallinity in agreement with the enlarged grain size shown in the SEM images above. Photoexcited carrier behavior was investigated using steady-state photoluminescence (PL). The steady-state PL spectra in Figure 4e showed that PL intensity for the perovskite film with MACl is over 15 times and 3 times higher than those of control and the one with NH_4Cl, respectively, indicating the suppressed non-radiative recombination. According to the Tauc plot in Figure 4f, The optical bandgaps were slightly affected by different

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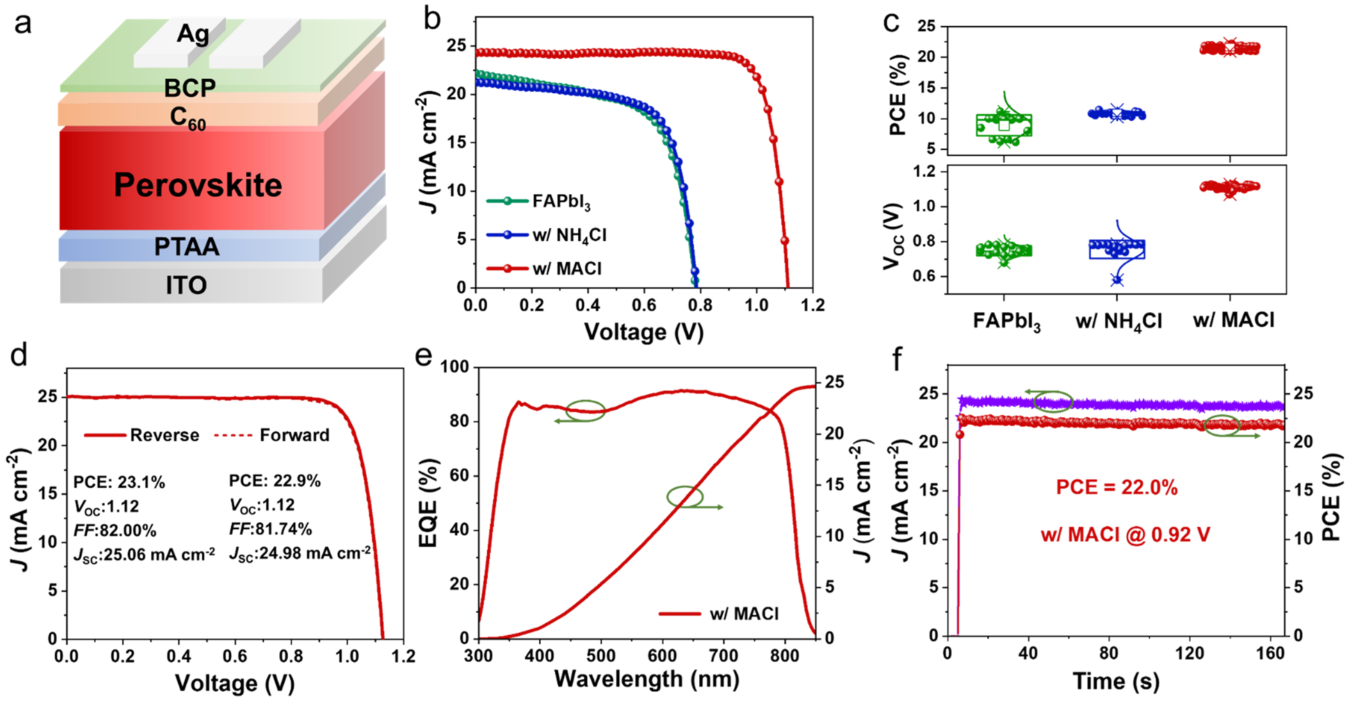

Figure 5. (a) Schematic device architecture of the inverted-structured PSC. (b) J−V curves of the PSCs based on $FAPbI_3$ for control, with NH_4Cl, and with MACl. (c) Statistical PCE and V_{OC} data of the corresponding devices. (d) J−V curve of the best-performance PSC based on the MACl additive. (e) External quantum efficiency (EQE) spectra of the $FAPbI_3$ PSC with the MACl additive. (f) Stabilized power output of the champion PSC tracked at the maximum power point (MPP) under AM 1.5G illumination.

volatile additives, consistent with that described in the literature.[52,53]

As shown in Figure 5a, the p−i−n-type inverted PSCs were fabricated with an architecture of ITO/PTAA/perovskite/C60/BCP/Ag to study the device performance of pure $FAPbI_3$ from three crystallization routes. Figure 5b shows the current density−voltage (J−V) curves of the optimized devices. The detailed photovoltaic parameters are summarized in Table S3. The reference device based on pristine $FAPbI_3$ showed an average PCE of 8.9% with a VOC of 0.75 V, a short-circuit current density (JSC) of 20.79 mA cm^{-2}, and an FF of 56.24% under one sun illumination (Figures 5c, S25, and Table S3). The FF showed a significant fluctuation of ±9.73%, which should be due to the pinholes of the pristine $FAPbI_3$ film from the SEM images above. The $FAPbI_3$ with NH_4Cl significantly increased the FF to 68.98 ± 5.69%, resulting in an enhanced PCE of 10.79 ± 0.32%. Furthermore, after the systematic device optimization, the MACl additive not only lowered the phase-transition temperature of $FAPbI_3$ but also achieved a higher film quality and undamaged interfaces, leading to an average VOC of 1.11 V and a PCE of 21.46%. To fully exploit the potential of the MACl additive, surface passivation and light management were further applied to improve device performance.[54] As shown in Figure 5d, the champion device based on $FAPbI_3$ with MACl achieved a PCE of 23.1% (V_{OC} = 1.12 V, J_{SC} = 25.06 mA cm^{-2}, and FF = 82.00%), which is the highest reported for $FAPbI_3$-based inverted PSCs (Table S1), demonstrating the great potential of MACl for enhancing device performance. To ensure data reliability, the external quantum efficiency (EQE) of the device was also measured to calibrate the cell JSC (Figure 5e). The integrated photocurrent with 24.65 mA cm^{-2} of the champion PSC matched well with the value obtained from the J−V measurement. A stabilized PCE of 22.0% for over 160 s was measured at a maximum power point for the MACl-based PSC (Figure 5f), agreeing well with the J−V sweeps. The operational stability measured under the maximum power point (MPP) condition in Figure S27 showed that the MACl-based device has better stability than that of the pristine device in a time span of 280 h.

■ CONCLUSIONS

In summary, this work systematically studied the functions of a series of Cl-based volatile additives and MA-based additives in $FAPbI_3$-based inverted PSCs. Using in situ photoluminescence, we provided clear evidence to unravel the different roles of NH_4Cl and MACl in the nucleation and crystallization of $FAPbI_3$. In addition, we found that MA-based additives (MACl, MABr, and MAI) could induce MA-rich nuclei to the α-phase $FAPbI_3$ compared to non-MA additives (NH_4Cl and FACl). Furthermore, volatile MACl could promote the growth of secondary crystallization during annealing. The optimized solar cell could achieve a PCE of 23.1%, which is among the best for $FAPbI_3$-based inverted PSCs. This work provides guidance for choosing appropriate additives for achieving highly efficient inverted PSCs.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/jacs.2c13566.

Material synthesis; device characterization; and stability measurements (PDF)

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Notes

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